

candidate $\tilde{c}_t$ contains the new signed content proposed for storage. The additive cell-state update provides a more direct path through time than the repeated non-linear replacement in a basic recurrent network. It supports long-term information transport, although the practically retained horizon still depends on the learned gate activations, training data, state dimension, and sequence handling.

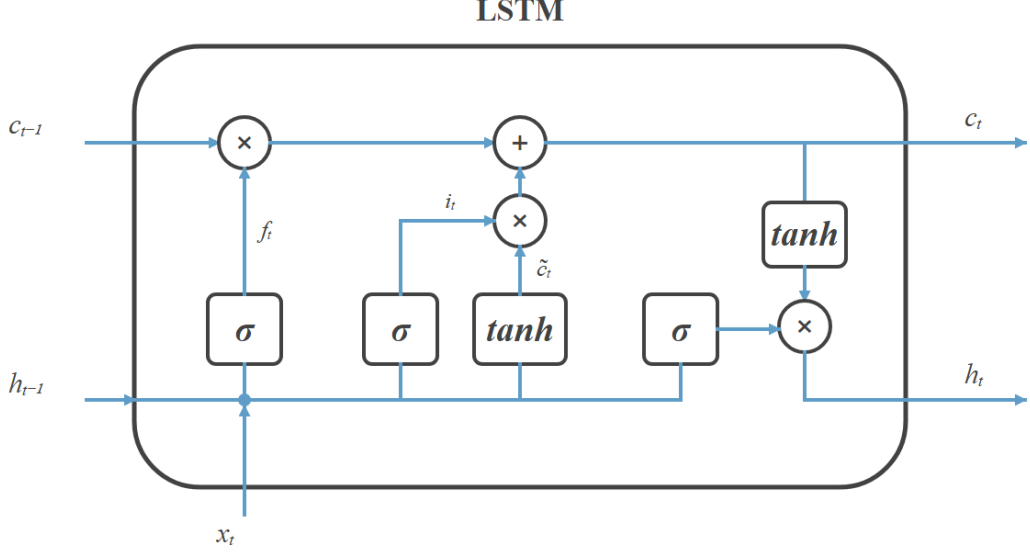


Figure 3.3: Detailed data flow through an LSTM cell corresponding to Equation (3.12). The forget gate controls the retained cell state, the input gate controls the candidate write operation, and the output gate controls the exposed hidden state. Blue paths indicate the directed state and gate signals, $\odot$ denotes element-wise multiplication, and the addition node forms the updated cell state.

Figure 3.3 separates the recurrent cell from any application-specific input preparation or output network. At each time step, the cell receives the current feature vector and both states from the preceding step. The updated hidden state can be passed to a regression head, while the hidden and cell states are carried into the next update. A unidirectional LSTM can therefore operate causally in an online BMS. A bidirectional network would additionally require future samples and is not suitable for the real-time estimators considered here.

For input dimension D and hidden dimension H , each of the four transformations contains an input matrix of size $H \times D$ and a recurrent matrix of size $H \times H$. With one combined bias vector, the parameter count is

$$N_{\text{param}}^{\text{LSTM}} = 4H(D + H + 1). \quad (3.13)$$